

ALEX CHEVRIER — Systems & AI Engineer

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SKILLS

Systems/HPC: C++23, Kotlin, Java, lock-free SPSC/MPSC, seqlock, DPDK, memory ordering, pthread_setaffinity_np

Hardware/AI: Vitis HLS, RTL co-simulation, FPGAs | Python, PyTorch, QLoRA, Local LLMs, Nomic Embeddings

OPEN SOURCE PROJECTS

Epistemic MLOps Engine (Python) | github.com/alchevrier/epistemic-filter

- Engineered framework distilling explicit hardware constraints (zero-allocation) into local edge LLMs.
- Built a two-axis evaluation gate matching vector domain relevance with structural first-principles reasoning.
- Implemented LLM-driven "Dynamic Discovery" to generate contrastive W-registers from architectural docs.

Autumn (Kotlin Multiplatform) | github.com/alchevrier/autumn

- Architected a circuit-based, zero-allocation frontend framework for deterministic mobile and web rendering.
- Bypassed JVM/native GC via flat byte-array registries, syncing raw socket payloads directly to Compose/SwiftUI via an epoch state engine.

FPGA Feed Handler (Vitis HLS/C++) | github.com/alchevrier/fpga-feed-handler

- ITCH 5.0 parser → AOS BRAM order book via a 4-stage DATAFLOW pipeline; dual-feed hardware arbitration.
- CoSim verified on Artix-7 (250 MHz): 13 cycles / 52 ns, II=16. 11 ADRs formally documenting constraints.

Low-Latency Feed Handler (C++23) | github.com/alchevrier/low-latency-feed-handler

- Designed zero-copy ITCH 5.0 parser → MPSC → SOA order book utilizing DPDK pcap PMD. Zero heap.
- Benchmarked 163M real NASDAQ messages: P99.9 at 274 ns parse + insert; 97.5% branch accuracy.

PROFESSIONAL EXPERIENCE

Morgan Stanley, Hong Kong — Software Engineer (Contract) (Nov 2025–Present)

- Architected greenfield messaging service; reverse-engineered Python batch system into streaming Kafka pipeline.
- Maintained core production Kafka infrastructure across isolated data centres.

We+, Hong Kong — Software Engineer (Sept 2023–Nov 2025)

- Spearheaded infrastructure migration for top-tier French investment bank, upgrading FX microservices to Java 21.

LandyTech, London — Software Engineer (Oct 2019–Aug 2023)

- Orchestrated monolith-to-microservices migration and fully automated robust risk data analytic pipelines.

Redbite Solutions, Cambridge — Software Engineer (Nov 2016–Oct 2019)

- Bootstrapped full-stack enterprise web applications; implemented secure OAuth2 authentication.

EDUCATION

ENSEIRB-MATMECA, Bordeaux — M.Sc. Computer Science (2013–2016)

Admitted by competitive national examination (Grande École entrance).